Stop at the red line

Name: ____________________

Date: ____________________ **Period:** ____________________

Today I can...

Make the robot move on white and stop on red or an unknown reading.

Words to know

If-then rule: If this happens, do that.

Loop: Steps that repeat.

Bounded: Set to stop after a certain time or number of steps.

LOOK AT THE IDEA

WHITE → **Move a little**

RED → **Stop**

OTHER → **Stop and check**

Gather your materials

- Checked base with downward-facing color sensor
- The tested white and red paper from lesson 4
- Removable tape and ruler
- Connected device with lesson 6 code

Build a red-line stopping station

Build with the power off. Tick each step when it is done.

1. Make a flat white lane at least twice as wide as the robot. Tape its edges down so the wheels cannot catch them. Keep tape away from the sensor's reading path.
2. Place a red strip across the full lane, starting about 10 cm deep in the direction of travel. This is a starting test size, not a guaranteed stopping distance.
3. Leave a large clear area beyond the red strip. Mark a start close enough that a short drive can reach it. Ask the teacher to check the whole layout.
4. With the wheels still, show the sensor white, red and another color. Fix wrong readings before driving. Keep the sensor gap from lesson 4.

Sketch the setup. Label the parts you will check.

Finish the setup

5. Build the program below. A variable is a value the program remembers. keepGoing remembers whether this test may continue.

6. At low speed, try one approach to red. Keep someone ready at Stop. When the program ends, measure how far the sensor moved past the start of the red strip.

7. Repeat three approaches. Separately test an unknown or other-color reading with wheels raised. After a red or unknown reading, show white again: the same test must stay stopped.

8. If it misses red, stop. Try a wider strip or a lower speed, one change at a time. Do not remove the 20-check limit or the Stop steps.

- [ ] Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

Show where your robot starts and how you will stop it.

Code it. Predict. Try it.

Make the program

1. Start stopped. Set keepGoing to Yes.

2. Repeat at most 20 times. Only if keepGoing is Yes: read the color.

3. If white: move slowly for 0.1 seconds, then stop. Otherwise: stop and set keepGoing to No.

4. After the repeats, stop and end. A person must start a new test.

Coding Canvas is the app where you make the program. Use your teacher-checked settings before a floor run.

keepGoing is a value the program remembers: Yes allows another check; No keeps the robot stopped.

Before you run: what do you think will happen?

After one try: what actually happened?

End of class 1: save the code, stop and power off, label your parts, and keep these pages.

Test it fairly

Name: ____________________

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Change speed OR strip width. Keep the light, sensor height, start and white paper the same.

A successful test

- ☐ Three approach runs stop when red is read.
- ☐ Red and unknown both end the test; white cannot restart it.
- ☐ The program ends after no more than 20 checks.

Record every result

Run	Saw red?	Distance past line (cm)	Stayed stopped?
1			
2			
3			

- ☐ Wheels raised: red and other/unknown readings stop it. Showing white again does not restart this test.

Change one thing. Try again.

The one thing we changed:

We kept these things the same:

Run	Saw red?	Distance past line (cm)	Stayed stopped?
1			
2			
3			

What could make the robot miss a narrow red strip?

Use a result: our change helped / did not help because...

Before leaving: save your code, stop and power off the robot, return parts, and hand in this module.